

Thinking Inside the Box: An Evaluation of An Novel Search-Assisting Tool for Supporting (meta)Cognition during Exploratory Search

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Exploratory searches involve significant cognitively demanding aiming at learning and investigation. However, users gain little support from search engines for their cognitive and metacognitive activities (e.g., discovery, synthesis, planning, transformation, monitoring, and reflection) during exploratory searches. To better support the exploratory search process, We designed a new search assistance tool called OrgBox. OrgBox allows users to drag-and-drop information they find during searches into “boxes” and “items” that can be created, labeled, and re-arranged on a canvas. We conducted a controlled, within-subjects user study with 24 participants to evaluate the OrgBox versus a baseline tool called the OrgDoc that supported rich-text features. Our findings show that participants perceived the OrgBox tool to provide more support for grouping and reorganizing information, tracking thought processes, planning and monitoring search and task processes, and gaining a visual overview of the collected information. The usability test reveals users’ preferences for simplicity, familiarity, and flexibility of the design of OrgBox, along with technical problems such as delay of response and restrictions of use. Our results have implications for the design of search-assisting systems that encourage cognitive and metacognitive activities during exploratory search processes.

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1 INTRODUCTION

Exploratory search tasks often involve learning new knowledge and investigating relationships among components of a topic. Exploratory searches are often ill-structured and involve open-ended goals. The search processes associated with exploratory tasks inherently involve uncertainty and ambiguity, as the searcher may not know what information they need or what they will find during the search process (Marchionini, 2006). Exploratory searches also frequently involve cognitively demanding activities including discovery, analysis, evaluation and monitoring (Marchionini, 2006; White & Roth, 2009; Wildemuth & Freund, 2012). Over the past decade, the information retrieval community has recognized that users need more assistance with exploratory search tasks compared to simple lookup tasks (Kelly, Arguello, & Capra, 2013; Harb, Sarnikar, & El-Gayar, 2015). However, current search engines do relatively little to support exploratory search (Culpepper, Diaz, & Smucker, 2018) and searchers are often left “on their own” to manage the collected information and their ideas using tools disconnected from the search system.

Although various research prototypes have been developed to assist users in discovering information and relationships among concepts, there have been few changes in how deployed search systems support users engaged in

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exploratory search tasks that involve learning and investigation. This has led to renewed calls for research about how search systems can support users engaged in task-based, exploratory, and complex searches where learning and exploration are goals (Culpepper et al., 2018). Researchers in the field of education and learning sciences have found that providing specific training and assistance to aid cognitive activities can lead to improvement of task performance and increased engagement during the e-learning process, which suggests that assistance tools can help users in learning and investigation. On the other hand, relatively few efforts in IR have focused on understanding users' cognitive activities and other mental processes that drive their interactions with systems and features during the exploratory search process. Our current study aims to fill this gap.

To provide support for exploratory searches and to better understand their underlying cognitive processes, we conducted a series of user studies. In an initial study, we investigated how users take notes and organize them during an exploratory task (Crescenzi, Li, Zhang, & Capra, 2019). Our findings revealed two primary search strategies (linear and grouping) and four primary information organizing structures (list, matrix, concept map, and article outline) used by participants. Drawing on these results and previous research on exploratory search, we designed a search support tool called the *OrgBox* to assist users with cognitive activities (e.g., organizing, synthesizing) and metacognitive activities (e.g., planning, and monitoring). The *OrgBox* tool is displayed in a separate window next to the search results and allows users to create "boxes" to organize information by dragging-and-dropping text from web pages into the boxes. Boxes can be given titles (headings) and personal notes can be added. Information can easily be re-arranged and indented using the drag-and-drop features.

Our research team conducted several studies to evaluate the *OrgBox* tool by comparing it to different baselines. In an early study (Choi, Arguello, Capra, & Ward, 2021), the *OrgBox* was compared to a bookmarking baseline tool similar to current Web browsers' bookmarking functions. The study found that the *OrgBox* helped users for exploring the task topics, organizing information, planning and monitoring their search processes more than the bookmarking tool.

We then conducted a second study with 24 new participants to compare the *OrgBox* to a Google Doc-style note-taking tool (called the *OrgDoc*) as the baseline. The *OrgDoc* mimicked features found in rich-text editors (e.g., bullet points, indentation, text formatting). For brevity, we refer collectively to the *OrgBox* and *OrgDoc* as the "OrgTools". In a previous paper (Crescenzi, Ward, Li, and Capra (2021)), we published results from this second study that focused on quantitative analysis of user interaction and user perception data. We observed that users viewed more articles, collected a greater amount of information, and used more information resources when using the *OrgBox* compared to the *OrgDoc*. Additionally, the quantitative results revealed that participants rated the *OrgBox* tool significant higher than *OrgDoc* both overall and in terms of providing more support for metacognitive processes such as monitoring coverage, task progress, and understanding the topic.

In this paper, we analyze additional data from the second study to gain a deeper understanding of the purposes behind users' interactions with the functions of OrgTools (such as dragging and dropping, adding, moving, deleting, and annotating) and to interpret their mental activities during the exploratory search process. In this paper, we analyze participants' responses from the retrospective interview that followed the search tasks. Through this analysis, we were able to gain insights into the ways in which users engaged with the OrgTools and how the tools supported cognitive and metacognitive processes that participants employed to learn about the task topics and to investigate relationships among the collected information. We report findings for the following three research questions:

RQ1: How does the *OrgBox* support users' cognitive activities during an exploratory search?

RQ2: How does the *OrgBox* support users' metacognitive activities during the exploratory search?

RQ3: How do users perceive the usability of the OrgBox compared to the baseline text editing tool?

By analyzing users' open-ended responses during interviews, our qualitative analysis was able to triangulate with our quantitative results, allowing us to extract more detailed explanations of users' perceptions of the OrgTools and gain a deeper understanding of users' cognitive and metacognitive processes during exploratory search. Our qualitative results offer a unique perspective compared to previous research – providing insights that help explain the cognitive and metacognitive purposes behind why users engage with certain features; emphasizing the importance of designing tools that support information behaviors during the exploratory search; and providing concrete explanations for what benefits users can gain from search scaffolding tools. Additionally, we present results about users' impressions of the usability of these features to help provide insights about the challenges that future designers should consider.

2 RELATED WORK

2.1 Exploratory search

During exploratory search, people go beyond searching for specific information and conduct investigation and examination to learn new knowledge and skills (White & Roth, 2009). Exploratory search is often associated with tasks that are open-ended, multi-faceted, and less-structured (White & Roth, 2009; Wildemuth & Freund, 2012). Users involved in exploratory searches are generally less familiar with the task topic domains and uncertain about their search strategies or task approaches (Marchionini, 2006; White & Roth, 2009). Consequently, exploratory search could involve various complex activities such as: comprehension, comparison, analysis, synthesis, planning, discovery, etc. Experimental studies found users frequently interlace the different activities through their search process (Marchionini, 2006; Bozzon, Brambilla, Ceri, & Mazza, 2013; Qu & Furnas, 2008).

The complexity associated with exploratory tasks and the search process brings up many challenges: searchers may need to identify search directions in an unfamiliar information space and formulate different but related queries (Klouche et al., 2015; Medlar, Ilves, Wang, Buntine, & Glowacka, 2016); navigate through an intricate information space (Sarrafzadeh & Lank, 2017; Dehghani et al., 2017); examine the relations among information collected from various sources (Golovchinsky, Diriye, & Dunnigan, 2012); and manage complex tasks at an actionable level (Kulahcioglu, Fradkin, & Palanivelu, 2017).

While not diminishing the importance of previous work, few studies have examined the connections between users' interactions with information and systems and their underlying cognitive processes. In addition, searchers receive limited support from current search engines during exploratory searches, often requiring them to use external tools or to rely on their own cognitive capabilities.

2.2 Cognitive and metacognitive activities

Much of the foundational research on cognition has been in the areas of psychology and education. Cognition is about the ways how our brain processes information—it refers to a range of mental activities relating to information perceiving, knowledge acquiring, retrieval, thinking, and reasoning (e.g., problem-solving, decision making) (Groome, 2021).

Researchers in the educational field developed a taxonomy of six cognitive skills associated with learning: remember, understand, apply, analyze, evaluate, and create, ranging from the simplest to the most complex (Anderson & Krathwohl, 2001). These cognitive processes function in a synergistic manner in their natural state (Groome, 2021). For instance, people categorize information based on their understanding of the similarities and differences. Meanwhile, categorization also provides bases for more complex cognitive activities such as applying, analyzing, and evaluating.

As a higher-order cognitive process, metacognition is about how people actively control their cognition (Watkins, 2001). It describes "the conscious self-awareness of one's own knowledge of task, topic, and thinking, and the conscious self-management (executive control) of the related cognitive processes." (Michalsky, Mevarech, & Haibi, 2009, p.364) (describing (Jacobs & Paris, 1987)). Metacognition consists of four central components for regulating one's cognitive process: 1) planning, 2) monitoring, 3) evaluating, and 4) revising (An & Cao, 2014).

Information-seeking tasks often include complex cognitive and metacognitive processes. In addition to Marchionini's (2006) lists of different types of cognitive and metacognitive activities in exploratory search, Toms (2015) suggests that people conduct extraction, analysis, comparison, prediction, and modification to complete exploratory tasks. And Klouche et al.(2015) noticed that exploratory search tasks involve cognitive attributes such as sense-making, decision-making, and problem-solving process. Conducting these activities during exploratory searches demands a great amount of effort beyond the accumulation of information. However, unlike research in the education area, relatively few efforts have specifically focused on supporting users' cognitive and metacognitive activities during an exploratory search. All of these findings suggest that exploratory searches call for support for users' cognitive and metacognitive activities.

2.3 Tools for supporting exploratory complex searches

Researchers from various areas and disciplines, including Human-Computer Interaction (HCI), information retrieval (IR), information organization, visualization, and education, to name a few, have developed prototypes and systems to assist users in complex task scenarios that involve exploring, identifying, and making sense of relevant information and knowledge acquisition. In the exploration and visualization areas, researchers have developed highly interactive interfaces to help searchers initiate searching and browsing for exploratory tasks and help them navigate through visualized spaces containing linked data sets or document entities. Examples include *Exploratory Wall* (Klouche et al., 2015) for mobile device exploratory search on touch screens, *KLink Search* (Shukla & Hoeber, 2021), *PULP* (Medlar et al., 2016) for exploring academic literature, and *hierarchical knowledge graphs (HKGs)* (Sarrafzadeh & Lank, 2017) developed to assist exploratory searchers in quickly identifying important concepts and planning further exploration of task topics. In addition, Dehghani et al. found that providing additional *browsing path recommendations* can facilitate more focused-searching during the exploratory search process.

Numerous tools have also been developed to support exploring, organizing, and note-taking activities in specific domains, such as document triage, educational, image, medical fields. Researchers observed that users often intertwine activities including searching, selecting, reading, and organizing to explore relevant documents from larger data sets (Bae et al., 2006). Visual spaces were developed to support users to conduct these activities more conveniently, enabling them to make quick relevance judgments regarding a document's relevance to their topic and relations among documents (Badi et al., 2006; Hearst & Degler, 2013). Zhang and Quintana (2012) designed *Digital IdeaKeeper* to help students plan, search, analyze, and synthesize information within the provided space. Stadler and Bromme (2008) developed *met.a.ware* to provide metacognitive prompting (e.g., evaluation, monitoring) and ontological classification to help layperson's searching for medical information. Liu, Medlar, and Glowacka (2022) designed a faceted search system *ROGUE* to help users explore and navigate spaces with images. Guo, Zhou, Gotz, and Wang (2023) developed the *GRAFS* interface to help exploratory searchers to construct their mental model of a topic area during an exploratory search.

Nonetheless, the evaluation of these tools has primarily concentrated on users' interactions with specific features. Few studies have explored the relationship between users' interactions with these systems and the underlying cognitive and metacognitive processes driving their actions. Our current paper aims to address this gap in the literature.

3 SYSTEM DESIGN

To explore our research questions, we built a custom search system and two search assistance tools – OrgBox and OrgDoc. The search system provided a standard search interface for the LDC New York Times corpus of over 1.8 million articles spanning the years between 1987-2007. Lucene was used as the back-end search engine.

The search interface allowed users to enter queries and would return results displayed in a standard search result page (SERP) format with a title, URL, and a query-biased text snippet. Clicking a result link would display a landing page with a text-only version of the corresponding NYT article. The interface displayed 10 results per page and included pagination controls at the bottom of the page to explore up to 50 total results. The system logged user interactions including clicks, mouse events, and queries. The search interface also displayed the task description at the top of the page and had a button to launch the OrgBox or OrgDoc (depending on the experimental condition) in a separate window to the right side of the main search window (see Figure 1).

We developed the OrgDoc tool (Figure 1) to provide a baseline condition that is similar to other text editors and online word processors (e.g., Google Docs or Microsoft Word). The OrgDoc provided a page that allowed users to enter and format information in a rich-text editor. They could type information directly into the OrgDoc, or could drag-drop and copy-paste text from landing pages (e.g., content, article titles, and URLs). The rich-text formatting features included: bold, italic, underline, multi-level indentation, and bullet items.

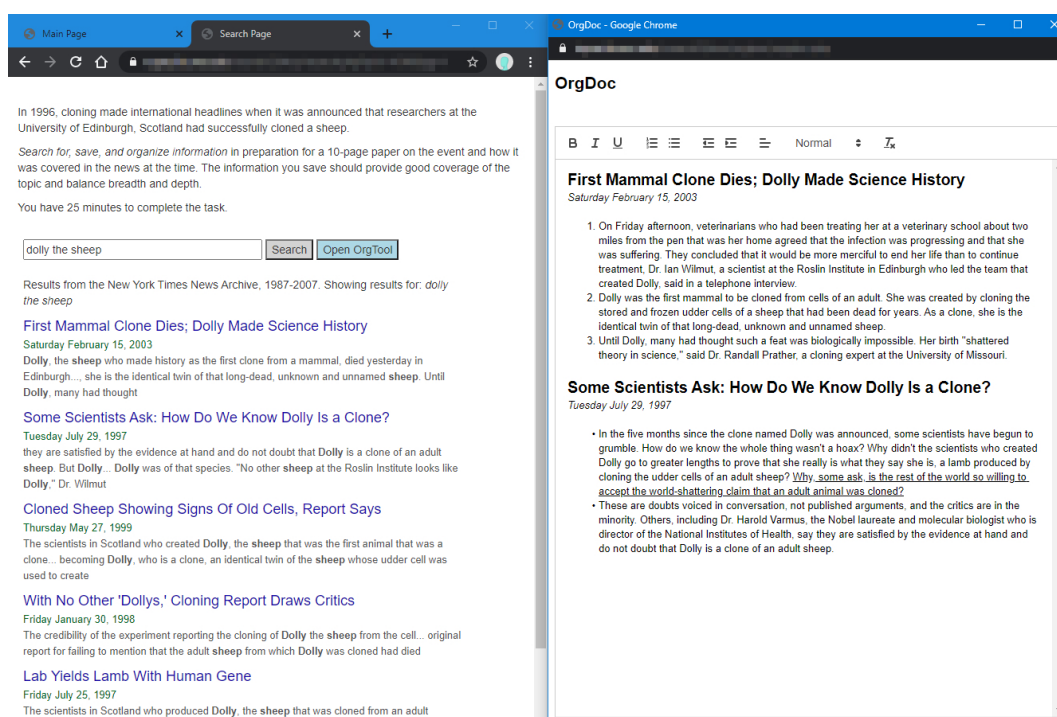


Fig. 1. View of the search system result page (left) and the OrgDoc tool (right) as participants were instructed to arrange their browser windows.

To assist users with various cognitive activities, we aimed to design a tool that facilitates easy information collection and storage, allows comparison and evaluation of collected data, supports organization reflecting users' thought processes during analysis, and helps users store creative ideas for new exploration directions. Additionally, the tool should provide metacognitive support, including functions to save information sources for tracking, and to provide a visual view to aid in evaluating task progress and adjusting search approaches.

With these objectives in mind, we designed the OrgBox tool (Figure 2) to provide participants with a means to collect, track, and structure information as they worked on exploratory search tasks. The OrgBox allowed participants to create "boxes" to act as containers for "information items" they found during their search. Participants could create an unlimited number of information items inside the boxes by dragging-and-dropping text from the landing pages (i.e. the NYT articles) into the OrgBox, or by clicking a "Manually Create Item" button at the top of each box (Figure 2, circle 2). The dragging-and-dropping text from the landing page into an OrgBox box would automatically copy over the text, page title, and URL and present the user with a dialog box to edit the content and/or add a note (Figure 3). Once saved, the information would appear in the box (Figure 2, circle 3). Participants could manually edit an item's text, title, or URL and could also add an optional annotation (note) to the item if they wished (Figure 2, circle 4). Items could be moved from one box to another, reordered in the same box, or nested into each other (Figure 2, circle 5) for a total of three levels by dragging and dropping the items. The boxes and items could be edited (by clicking the wrench icon) or removed (by clicking the X button). Participants could also add titles to boxes and re-arrange the position of the boxes.

Both the search system and the OrgTools were instrumented to allow us to capture interaction data (mouse movements, text drag-and-drops, copy-and-pastes, use of formatting options) as participants completed their tasks¹.

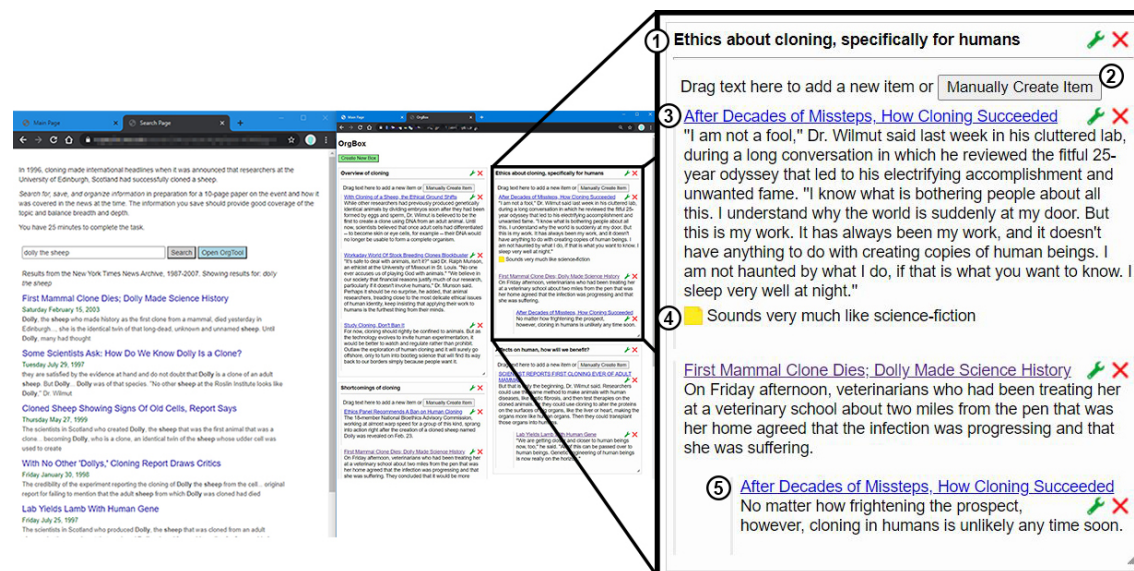


Fig. 2. View of the search system result page (left) and OrgBox tool (center, right). The OrgBox displays (1) box title, (2) "Manually Create Item" button, (3) item title, URL (as a link), edit and remove buttons, and text, (4) item note/annotation, (5) nested items

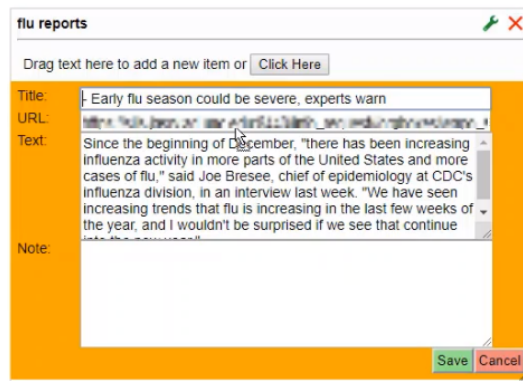


Fig. 3. Item Save Dialog window that appears after the text has been dragged and dropped into a box in the OrgBox.

4 METHOD

We conducted a within-subject user ($n=24$) study in which each participant completed two exploratory search tasks, one using the OrgBox and one using the OrgDoc. We used complete sets of counterbalanced topic-interface combinations to balance the study design. Participants were recruited through the university's email mailing lists and from a pool of undergraduate students, graduate students, and university staff. We recruited 14 female and 10 male students between the ages of 18 and 35 ($M = 23.8$, $SD = 5.08$). Since the study was conducted remotely using video conference software, participants were screened for access to a computer with a minimum display resolution of 1200x800. This study was approved by the Institutional Review Board (IRB) at the University of North Carolina at Chapel Hill.

4.1 Tasks

Each participant in our study completed two exploratory search tasks – one using the OrgBox and one using the OrgDoc (counterbalanced). For each task, we gave participants the same simulated work task scenario (writing a 10-page paper) with a different task topic. The two tasks were completed separately – e.g., participants completed the scenario with one task topic and then completed the scenario with the second task topic (counterbalanced).

During the study, we asked participants to search for information for each task and use the OrgTools to save and organize the information they found to complete each task. However, since the focus of this study was on the search process, we did not ask participants to actually write the 10-page paper. The work task scenario, task topics, and instructions are shown below.

Scenario: Imagine that you are taking a class that requires you to write 10-page papers on significant world events and how these events were covered in the news in the past.

Task 1: In 2005, Hurricane Katrina struck the United States Gulf Coast causing massive destruction and great loss of life.

Task 2: In 1996, cloning made international headlines when it was announced that researchers at the University of Edinburgh, Scotland had successfully cloned a sheep.

Task instructions: Search for, save, and organize information in preparation for a 10-page paper on the event and how it was covered in the news at the time. The information you save should provide good coverage of the topic and balance breadth and depth. You have 25 minutes to complete the task.

4.2 Protocol

The study was conducted as a remote, synchronous study using the Zoom video conferencing platform. Each study session began with an introduction by the moderator that included: reviewing and completing an informed consent form, giving instructions about how to use Zoom screen sharing, a display resolution check, and a practice task that introduced the search system, the drag and drop interaction, and the think-aloud protocol.

Participants completed the two tasks following the same steps. For each task, the participant was asked to view the scenario, task topic, and instructions. Then they were asked to complete a pre-task questionnaire and watch an introductory video describing the OrgTool they would be using for the task. After the video, the participant was instructed to open the search page and begin the task. The moderator also reminded the participant to think aloud as they searched. Participants were given up to 25 minutes to work on each task. The task was considered finished after the participant signaled that they had completed their task goals or if they reached the 25-minute time limit. Then the participant would complete a post-task questionnaire about their perceptions of the search system and the OrgTool. This set of steps was then repeated for the second search task. After completing the post-task questionnaire for the second task, the participant completed an exit questionnaire which consisted of four open-ended questions asking what they liked and disliked about each system. On completion of the exit questionnaire, the moderator conducted a semi-structured interview with the participant in which they were asked about the OrgBox features they interacted with, why they used (or not) a specific feature, and the feature they like most or dislike most. Finally, the participant was compensated with a \$40 USD Amazon gift card for their efforts.

The search processes along with participants' think-aloud comments for both tasks were recorded using the Zoom video conference recording feature. The interview was audio-recorded including a screenshot of the information, the user collected/created in the OrgTool. The entire study lasted about 2 hours.

4.3 Data collection and analysis

4.3.1 Logged interactions. The system logged all participants' interactions with the search interface (e.g., queries, clicks, scroll events, and drag-drop events). The system also logged all interactions with the OrgTools including creating boxes/items, editing boxes/items, changing positions boxes/items, formatting changes, and deletions.

4.3.2 Questionnaires. Participants completed pre-task and post-task questionnaires before and after each task, and an exit questionnaire at the end of the session. The questionnaires included agreement scale questions (7-point scales; 1=strongly disagree, 7=strongly agree) and open-response questions to gather data about participants' perceptions and impressions. The *pre-task questionnaire* included six items about their interest in the task, prior knowledge about the task topic (2 items), and the perceived difficulty of the task (3 items). The *post-task questionnaire* included items about: perceived cognitive and metacognitive benefits of the OrgTools (13 items); task difficulty (3 items); workload (6 items from the NASA-TLX (Hart & Staveland, 1988)); participants' perceptions of the quality and coverage of the collected information they created using the OrgTools (4 items); and perceptions of the usability of the OrgTools (4 items) and the search system (4 items). The *exit questionnaire* contained four open-response questions that asked participants to describe things they liked and disliked about each OrgTool.

4.3.3 Data analysis. To address our RQs, we conducted both quantitative and qualitative analyses.

Quantitative analysis – Our quantitative analysis focused on (1) the logged search activities (i.e., number of queries, number of articles viewed), (2) participants' interaction with the OrgTools (i.e., amount of text dragged into the OrgTool,

number of sections created), and (3) participants' perceptions of the OrgTools cognitive and metacognitive benefits from the post-task questionnaire. Table 1 summarizes these measures.

Qualitative analysis – We conducted qualitative content analysis on participants' responses to the open-ended questions from the post-task questionnaire and the exit questionnaire. The process to open code the responses were as follows. First, two of the authors independently went through a subset of participants' responses to each create their own set of initial codes (open coding). The coders met and discussed all the codes identified until a combined initial set of codes was agreed upon. For the second round of coding, each coder used the initial codebook to code another subset of the data and add new codes if any arose. A second meeting was held to discuss how well the initial code set worked and to incorporate any new codes that had emerged. Each coder then used the new code set to independently code all responses (including the first two subsets) for all participants. If needed, each coder also added new codes. The two coders held a final meeting to work through all the coding disagreements until a complete agreement was achieved. The purpose of this iterative coding process was to make sure that we captured as many of the important aspects mentioned by participants as possible. Using the same open coding approach, the two coders also analyzed the collected information and notes that participants created using the OrgTools during their search processes. The analyses of this information and notes were also consulted during the data analysis stage to help understand participants' usage of different OrgTools' functions.

5 RESULTS

In this section, we first summarize results from our quantitative analysis of the logged interaction data and participants' responses to the agreement scale questions². Next, we present results from our qualitative analysis to address the main research questions posed in this paper: How does the OrgBox support cognitive (RQ1) and metacognitive (RQ2) activities during the search? And how do participants perceive the usability of the OrgBox compared to the baseline text editing tool (RQ3)?

5.1 Quantitative results

Table 1 summarizes key results from our quantitative analyses of the logged interaction data and participants' responses to the post-task questionnaire agreement scale questions. Four main categories of measures are presented in Table 1: (a) interactions with the search system, (b) interactions with the OrgTools, (c) participants' perceptions of the OrgTools and search system, and (d) participants' perceptions of the cognitive and metacognitive support provided by the OrgTools. Results for each of these are presented below.

Interactions with the search system – Participants issued more queries ($OB_M=5.46$, $OD_M=4.08$), and viewed more articles ($OB_M=11.54$, $OD_M=9.83$) for tasks using the OrgBox versus the OrgDoc, but these differences did not reach statistical significance. Meanwhile, participants viewed significantly fewer articles per query with the OrgBox condition compared to that of the OrgDoc condition ($OB_M=2.50$, $OD_M=4.05$). The results suggest that as the search system and OrgTool did not affect users' search activities significantly, participants issued more queries and examined more articles for different queries using the OrgBox instead of reading more articles for the same query when using the OrgDoc.

Interactions with the OrgTools – We found several significant differences related to participants' interactions with the OrgTools. First, participants did significantly more dragging and dropping of text with the OrgBox compared to the OrgDoc ($OB_M=20.21$, $OD_M=13.50$). Second, participants added more words per drag/drop (or copy/paste) using the OrgBox versus the OrgDoc ($OB_M=80.66$, $OD_M=54.91$). Third, they saved/added significantly more article titles per task when using the OrgBox compared to the OrgDoc ($OB_M=7.88$, $OD_M=3.00$). This third result is not surprising

since the OrgBox automatically saved the title and URL of each article when the drag-drop feature was used, but in the OrgDoc this information had to be entered manually. Fourth, participants created significantly more section groupings when using the OrgBox versus the OrgDoc ($OB_M=5.25$, $OD_M=4.04$). In our analysis, we counted sections as a block of information that was separated from other information. In the OrgDoc, a section could be a paragraph, a group of bullet points, or related text separated by blank lines, space, or another formatting. In the OrgBox, we considered each box created to be a separate section. Fifth, participants saved more words in their final collection of information when using the OrgBox compared to the OrgDoc ($OB_M=1163.04$, $OD_M=966.71$). Across all these interaction measures, it is apparent that participants saved more information and created more sections using the OrgBox compared to the OrgDoc tool.

Table 1. Participants' interactions with the search systems and OrgTools, their perceptions of OrgTools, the quality of the collected information and notes in the OrgTools, as well as their perceptions of cognitive and metacognitive benefits (1-7 Likert scale) by OrgTool conditions (mean, SD).

Measure (items or composite variables)	OrgDoc M (SD)	OrgBox M (SD)	Contrast χ^2 (1)
Interactions with the search system			
Number of queries per task	4.08 (3.00)	5.46 (3.35)	2.77
Number articles viewed per task	9.83 (3.36)	11.54 (6.87)	1.86
Number of articles viewed per query	4.05 (3.67)	2.50 (1.34)	* 4.44
Interactions with the OrgTools			
Number of text drags	13.50 (10.76)	20.21 (7.47)	‡ 16.72
Number of words per add	54.91 (47.75)	80.66 (52.76)	‡ 12.98
Number of article titles	3.00 (4.36)	7.88 (5.59)	‡ 7.14
Number of sections created	4.04 (2.27)	5.25 (1.62)	* 4.81
Number of words in final collection	966.71 (684.32)	1163.04 (672.19)	* 5.35
OrgTools and search system perceptions			
Grade (0 -100) for the OrgTool system	75.50 (16.01)	85.58 (9.49)	† 7.23
Grade (0 -100) for the Information added to the OrgTool system	77.63 (10.52)	78.25 (12.34)	0.05
Search system usability perceptions (composite of 4 questions, $\alpha=.69$)	5.84 (.81)	5.64 (1.03)	1.67
Perceptions of OrgTool's cognitive and metacognitive supports (composite variables)			
Support for monitoring/tracking notes coverage (composite of 3 questions, $\alpha=.73$)	4.76 (.97)	5.75 (.84)	† 13.86
Support for metacognitive evaluation of the process, task progress (composite of 5 questions, $\alpha=.80$)	4.38 (1.12)	5.10 (.93)	† 10.32
Support for understanding task topic (composite of 2 questions, $\alpha=.86$)	4.33 (1.20)	5.31 (1.37)	* 6.85
Support for metacognitive planning (composite of 3 questions, $\alpha=.72$)	3.69 (1.03)	4.17 (1.38)	4.00
Average of all OrgTool support questions (composite of 13 questions, $\alpha=.86$)	4.30 (.78)	5.07 (.81)	† 12.79

Note: Cronbach's α was used for composite variables; the 1-7 Likert scale was used for measuring perceptions on search system usability, and cognitive and metacognitive support questions where 1=strongly disagree, 7=strongly agree. The test statistic for contrasts of marginal linear predictions of OrgBox vs. OrgDoc conditions after multilevel mixed-effects negative binomial models (χ^2 (1)). * $p<.05$, † $p<.01$, ‡ $p<.001$.

Overall perceptions – In the post-task questionnaires, we asked participants to grade the OrgTool system that they had just used on a scale from 0 to 100. Participants graded the OrgBox significantly higher than the OrgDoc ($OB_M=85.58$, $OD_M=75.50$). We also asked participants to give a grade (0 to 100) to the information they had just gathered for the task. For this measure, we did not observe a significant difference between the OrgBox and OrgDoc tools ($OB_M=78.25$, $OD_M=77.63$). We also asked participants four questions about their perceptions of search system's usability (on a 7-point scale). These questions were specifically focused on the search system (e.g. it provided relevant results; results were easy to filter), not on the features of the OrgTools. We did not observe a significant difference in this score for the search system between the OrgBox and OrgDoc conditions ($OB_M=5.64$, $OD_M=5.84$).

Perceptions of OrgTool's cognitive and metacognitive support – Based on a quantitative exploratory factor analysis of 13 metacognitive support items³, our results show that participants reported significantly higher levels of support for both cognitive and metacognitive activities when using the OrgBox compared to the OrgDoc. Specifically, the OrgBox

was perceived to provide significantly more support for monitoring/tracking notes coverage ($OB_M=5.75$, $OD_M=4.76$), evaluation of the search and task progress ($OB_M=5.10$, $OD_M=4.38$), and understanding the topic ($OB_M=5.31$, $OD_M=4.33$). Although it did not reach statistical significance, participants rated the OrgBox higher than the OrgDoc in terms of planning ($OB_M=4.17$, $OD_M=3.69$). In addition, there was also a significant difference in the average across all 13 questions with the OrgBox having a higher score than the OrgDoc ($OB_M=5.07$, $OD_M=4.30$). These results provide strong evidence that participants thought the OrgBox gave more metacognitive support during an exploratory search compared to the baseline OrgDoc.

5.2 RQ1: OrgBox support for cognitive activities

The above quantitative results provide strong evidence that participants viewed the OrgBox as providing more support for cognitive and metacognitive activities during the exploratory search. Next, we address *how* the OrgBox provided this support for cognitive (RQ1) and metacognitive activities (RQ2). To investigate this, we conducted qualitative analysis of participants' responses to the open-ended questions from the post-task questionnaire and exit questionnaire.

As noted earlier, cognitive activities that are important during exploratory searches include information extraction, understanding, analysis, comparison, planning, and synthesis (Marchionini, 2006; Toms, 2015). Our analysis identified five specific ways that the OrgTools provided cognitive support, including creating and maintaining structure, grouping information, re-positioning information, formatting text, and annotating (RQ1). Each of these is discussed below.

5.2.1 Creating/maintaining structure. Creating and maintaining a structure of information encountered involves multiple cognitive activities including identifying information, extracting important concepts, and understanding the relations between concepts through interpreting, comparing, and organizing (Anderson & Krathwohl, 2001). These activities are connected to users' interactions with the search system and the information returned through actions such as issuing queries, identifying relevant articles, reading articles, creating sections in the OrgTools, and dragging/copying text to the OrgTools. Participants described how they used features of the OrgBox and OrgDoc to capture and externalize the structure of information they encountered. These included the need to create, maintain, and repair the structures as they encountered new information during the search process. Both the OrgBox and OrgDoc had features that could be used to save, structure, and organize information. For example, when using the OrgDoc, participants noted how they used the text-formatting features to create divisions and structure:

"I used bold to make the article names stand out, underline to organize by category, and italics to differentiate my personal notes from the quotes I pulled from the articles." (p2-OrgDoc)

When using the OrgBox, participants noted how the built-in features of the boxes (e.g., the visual frames around boxes, the required heading field, and the ability to create hierarchical levels) helped them structure and organize the information they were found:

"I liked how the system [OrgBox] required me to create headings, encouraging me to conceptualize sub-topics. I also liked how there was a built-in nesting system." (p6-OrgBox)

Participants noted that the OrgDoc required more effort to create the structure (e.g., by using different formatting styles) than using OrgBox:

"[Using OrgDoc, I] Had to create own sections/subtopics to organize thoughts where OrgBox did that for me." (p16-OrgDoc)

Table 2. Number of participants for perceived benefits and challenges of the OrgBox contrasting to the OrgDoc.

categories		OrgDoc (#)	OrgBox (#)
Support for cognitive activities			
Benefits	creating, maintaining structure	7	9
	grouping information	-	5
	re-positioning information	-	14
	annotating	5	6
	formatting text	12	-
Challenges	lack of structuring help	3	-
	missing formatting features	1	5
Support for metacognitive activities			
Benefits	supporting information source track	-	10
	supporting thought process	-	5
	supporting task progress track	-	2
	assisting visual overview	1	9
Challenges	lack of source tracking	9	-
	lack of support for thought process	4	-
	more planning required	2	-
OrgTool usability evaluation			
Benefits	simplicity	9	5
	familiarity	11	2
	flexibility	9	1
	drag and drop	4	7
	keyboard function	4	-
Challenges	loss of text formats	4	3
	issues with drag and drop	3	6
	need more space	4	3
	cumbersome / restrictions	3	12
	unexpected box movements	-	4
	redundant source tracking	-	2
	lack of OrgBox functions	5	-

5.2.2 Grouping and re-positioning information. Grouping and re-positioning information reflects participants' cognitive processes of classifying, comparing, differentiating, organizing, and summarizing (Anderson & Krathwohl, 2001). These cognitive processes are embodied in observed behaviors such as creating new sections, dragging text to different sections, merging sections, and moving items among sections. Participants mentioned the OrgBox support for separating information into groups, merging information from different groups together, and nesting information hierarchically by dragging and dropping. Participants also noted how the grouping and re-positioning features in the OrgBox supported a natural, non-linear method to organize the found information:

"I liked that [OrgBox] allowed me to bring over information quickly and organize by block, instead of just scrolling text. I also appreciated the not-completely-linear flow, with the multiple columns because I could organize ideas more fluidly." (*p3-OrgBox*)

5.2.3 Annotating. Annotating is an important cognitive activity that helps users construct meaning and interpret the retrieved information through clarifying, paraphrasing, inferring, even critiquing (Anderson & Krathwohl, 2001). In the OrgBox, participants could use the annotation function to add annotations to each information item saved in a box. But in the OrgDoc, participants could add notes at any place and format the notes manually. We found several

different types of annotations participants created: (1) metadata about the collected information, such as dates and source information; (2) perceptions, interpretation, and interesting points about the saved information; (3) summaries of the context (the original article) where the information was extracted; (4) descriptions about relationships (e.g., similarity, differences, connections) with other information/ concepts/ boxes; (5) ideas and reminders (e.g., about new possibilities for search directions and ideas about generating the final outcome).

By comparing the annotation types included in both the OrgBox and OrgDoc tasks, we found that annotations added into the OrgBox covered all of these types, but most annotations created in the OrgDoc were metadata information such as dates and URLs. This might be because 1) the OrgBox automatically copied over metadata such as the page title and URLs when the text was drag-dropped from a landing page into a box, 2) OrgBox displayed a dialog note box that included an empty field that prompted users to review and add notes to the saved information. (Figure 3). Whereas the OrgDoc tool lacked this stimulus and required more effort to create and format annotations.

“The “note” feature allows me to create a brief annotation about a huge chunk of text, which allows me to get an idea of the large chunk of text without having to read through all of it.” (*p9-OrgBox*)

“I used...italics to differentiate my personal notes from the quotes I pulled from the articles” (*p2-OrgDoc*)

5.2.4 Formatting text. Formatting text involves cognitive activities such as differentiating, selecting, organizing through outlining, and creating structures (Anderson & Krathwohl, 2001). We primarily observed these activities through participants using the hierarchical nesting function in the OrgBox and the text formatting features in OrgDoc. Participants commonly used text formatting to indicate the structures and relationships among their collected information items. Formatting information was also used to help make the content more organized and easier for reading.

The OrgBox provided only one method for formatting: items in a box could be nested up to three levels deep (Figure 2, circle 5). On the other hand, the OrgDoc contained multiple different formatting features (e.g., bold, underline, add bullet points) that allowed participants to style the text in the OrgDoc. Although the nesting feature of the OrgBox was useful to create a hierarchical structure, many participants described how the text formatting feature in the OrgDoc was helpful and familiar.

“The [OrgBox’s] ability to put information in a hierarchy allowed me to think more logically about the organization of the information.” (*p9-OrgBox*)

“The text formatting options of the OrgDoc system allow me to format my notes in a way that is clear to me. I also like how similar it is to note-taking applications that I am already familiar with, such as Google Docs or Microsoft Word.” (*p10-OrgDoc*)

However, participants also noted that manually formatting text in the OrgDoc could cause extra cognitive work and slow down the task speed. For instance,

“The formatting options [of OrgDoc] were great but meant that it took a while/many clicks to get the text where I wanted. It detracted from the content of the task for sure when I was frustrated with the tab not working to move my bullet point, for example. I felt a little more aimless with OrgDoc than OrgBox in terms of what categories to establish from the text.” (*p5-OrgDoc*)

5.3 Support for metacognitive activities

While there are many forms of metacognition, in RQ2 we focus on participants’ awareness of their own thinking process and the regulation of cognition such as monitoring, evaluation, and planning. Our qualitative analysis revealed four

ways that the OrgBox provided support for participants' metacognitive activities: (1) support for tracking information sources, (2) support for thought processes such as planning and reflecting, (3) support for keeping track of progress toward goals, and (4) assisting visual overview.

5.3.1 Support for tracking information sources. The OrgBox automatically saved source information (titles, URLs) when participants dragged and dropped a piece of information into the OrgBox. Tracking information sources is part of users' metacognitive process of monitoring. Our quantitative analysis found that compared to OrgDoc, OrgBox significantly helped users to monitor the diversity of information sources covered in their collection. This included evaluating the breadth of the topic covered and to easily trace back to the original articles when needed. Our qualitative analysis of participants' interview comments revealed how this feature of the OrgBox helped their metacognitive process of tracking (e.g., tracking sources and facets).

"It was easy to tell what facets I was covering, to compare them to each other, and to see how much information I had in each box. (*p2-OrgBox*)

"I liked that the OrgBox tracked your sources which would make it easy to do a "Works Cited" page and also to find more info from the article if you needed it later on." (*p23-OrgBox*)

Conversely, the OrgDoc did not offer automatic support for monitoring information sources, and participants had to consciously remember to save and organize information by sources. This resulted in an additional metacognitive burden for monitoring information sources of the collected information. For instance,

"[The OrgDoc] did not automatically pull in source, URL, article title, etc. [I] had to create own sections/subtopics to organize thoughts where OrgBox did that for me. [It was] Hard to know what notes came from which source once multiple articles were pasted in" (*p16-OrgDoc*)

5.3.2 Support for thought processes such as planning and reflecting. Planning and reflection are important metacognitive activities that involve understanding task requirements, developing plans to achieve goals, and adapting strategies as necessary (Hacker, Dunlosky, & Graesser, 1998). Participants described how the box feature of the OrgBox helped them learn more about their own thoughts, identify gaps, generate ideas for searching for new information, and organize their thoughts in the process. For instance,

"[the OrgBox]... allowed me to have a clearer mental picture of the information. The ability to put information in a hierarchy allowed me to think more logically about the organization of the information." (*p9-OrgBox*)

"[OrgBox is more] easier to organize my train of thought, save quotes that grab my attention" (*p18-OrgBox*)

In contrast, participants mentioned that the OrgDoc tool did not provide the same support to organize ideas and thoughts but required more planning to use.

"[the OrgDoc] requires a game plan, whereas OrgBox allows for your thoughts to be all over the place and it organizes them as you go" (*p18-OrgDoc*)

5.3.3 Supporting task progress tracking. Tracking task progress is another important part of metacognitive processes during exploratory search. This type of metacognitive tracking helps participants' awareness of their level of knowledge as they make progress, enables them to determine when they have achieved task goals (Hacker et al., 1998), and allows them to regulate their exploration process. Our qualitative analysis illustrated ways that participants found the OrgBox helpful in tracking and assessing the breadth and depth of their collected information within the boxes. They

specifically noted that it was easy to monitor the number of aspects (i.e., box numbers) and the amount of information (i.e., individual box size) distributed among different aspects.

“[the boxes] made it easy to keep track of what info belonged in which box and to compare how much info you had for each topic.” (p23-*OrgBox*)

On the other hand, some participants noted that OrgDoc was not effective in tracking their task progress, and it became increasingly challenging to organize and use the information as more data accumulated during the search process.

“I experienced difficulty with OrgDoc... my information became disorganized and even unusable at some point.” (p2-*OrgDoc*)

5.3.4 Assisting visual overview. Visually thinking can facilitate cognitive and metacognitive activities, especially for monitoring, planning, and evaluation (Kirsh, 2005). Participants noted that visual cues of the OrgBox features (i.e., number, size, position) offered quick insights into their thought process, helping them to determine how to organize the information and adjust their approaches to the tasks.

“The main advantage of the OrgBox system in my opinion is that it provides a very clear and visual way of organizing thoughts that may be subject to quick changes because the outline is still very much in a planning phase.” (p10-*OrgBox*)

The boxes also facilitated quick testing (i.e., evaluation) of ideas during brainstorming sessions.

“I liked the visual aspect of the boxes and sections and fluidity of moving those sections... which I like to use to organize thoughts when brainstorming” (p20-*OrgBox*)

Additionally, visual aids can be particularly helpful for individuals who are visual learners and prefer to absorb information through visual means.

“OrgBox appeals to the visual thinker. I liked to see the boxes/categories next to each other. By dragging the text to create subpoints within a category I was able to see the frame of the eventual paper better with OrgBox.” (p5-*OrgBox*)

5.4 Evaluation of OrgTools' usability

We were also interested to learn about users' perception of the usability of the OrgBox as a novel tool (RQ3). We found that participants' positive evaluation of the OrgTools' usability focuses on simplicity, familiarity, flexibility, and functions like drag-and-drop. In addition, they also pointed out some difficulties and challenges of use as well as suggestions for future improvements.

5.4.1 Simplicity, familiarity, and flexibility. Participants commented that both the OrgBox and the OrgDoc were simple and easy to use (OB=5, OD=9). For instance,

“[I like OrgBox for] how fluid it the experience was, how easy it was to create new boxes, and how many simple yet useful features it had (like the notes section).” (p24-*OrgBox*)

“Its [OrgDoc] simplicity and familiarity allow users to pick it up instantly without requiring much thought or effort.” (p9-*OrgDoc*)

Unsurprisingly, compared to the OrgBox, more participants (OB=2, OD=11) noticed that the OrgDoc was more similar to other text-editing tools such as the formatting functions, the space for saving and editing information and being able to use keyboard functions like shortcuts.

“This system [OrgDoc] is very close to what I usually use to take notes (Evernote). Because I am familiar with it, I was able to organize my notes how I wanted.” (p7-OrgDoc)

5.4.2 *Drag-and-drop text for information organization.* Many participants thought the *drag and drop text* feature was intuitive to use in both OrgBox and OrgDoc (OB=7, OD=4). Specifically, they liked that the drag and drop function in the OrgBox could help them quickly organize the collected information: besides using it as the copy and paste function, participants could drag and drop information into an existing box as a *new item* to previously created sections, or they could drag and drop the information into the blank area to create a *new section (box)* and immediately edit it by adding the section title, annotations and move the new section around to reflect its relationship with other sections.

“The ability to drag-and-drop text is very intuitive and convenient.” (p9-OrgDoc)

“the drag-and-drop system allowed me to bring over information quickly, and organize by block, instead of just scrolling text. I also appreciated the not-completely-linear flow, with the multiple columns, because I could organize ideas more fluidly.” (p3-OrgBox)

5.4.3 *Challenges of use and suggestions for improvements.* Participants also reported some challenges with using the OrgTools. Some issues were common for both OrgBox and OrgDoc, such as loss of text formats when dragging and dropping the information from landing pages into the OrgTools, system delays and the *drag and drop* function did not work or not reacting immediately, and more space was needed as the collected information is overfull, etc.

“When I tried to cut and paste some text, the formatting was reset and I did not know how to get my formatting back.” (p10-OrgDoc)

“The highlight feature that allows you to capture text is frustrating because it occasionally took me several tries to get the text extracted.” (p21-OrgBox)

As a novel tool, OrgBox was more attractive to use than OrgDoc for its novelty. Several participants (n=5) commented that OrgDoc was “no different from any other word processing programs” (p19-OrgDoc). Meanwhile, there were several important issues with the OrgBox. First, it was more cumbersome and restrictive for use compared to the OrgDoc. The OrgBox requires more effort to click. Secondly, since every box was designed to adjust its size and position based on the amount of content in it, sometimes unexpected box movements happened and caused confusion. Besides, the drag and drop function creates redundant links for each information item even when they are from the same information source.

“I didn’t like that the boxes seemed programmed to move based on the amount of text I added, ...I couldn’t find the section I wanted to add text to as it had moved.” (p20-OrgBox)

“I disliked that the article titles were on every note I dragged in and there weren’t options to change the text” (p22-OrgBox)

To address some of these problems, participants provided improving suggestions for OrgBox in their responses. These suggestions focused on two aspects: (1) providing more functions for editing the collected information and formatting the structure in the OrgBox. For instance, allowing users to resize boxes and items in a box, add formatting functions like colors, allowing to add more hierarchical levels to nest items, track editing changes, etc. (2) provide options to control certain functions manually. For example, allowing users to turn on and off the reflowing/repositioning of boxes. These suggestions could be taken into consideration for the future design of OrgBox.

6 DISCUSSION

In this paper, we presented findings on how our OrgBox search assistance tool can support users from a cognitive and metacognitive perspective (compared to the OrgDoc text-editor baseline tool). We summarized and presented qualitative analysis of users' self-explained information behaviors using the OrgTools during exploratory search tasks. In the following section, we triangulate our qualitative findings with: 1) our previously published quantitative findings about the OrgBox (Crescenzi et al., 2021), and 2) results from other prior research. By doing so, we discuss the connections and distinctions between our work and the existing literature.

RQ1: Supporting cognitive activities Our qualitative analysis found that OrgBox provided valuable support for participants' cognitive activities during the search process. Compared to OrgDoc, using the OrgBox was especially convenient for creating structures, grouping and re-positioning information by dragging and dropping items into boxes and modifying the positions of information items as they searched. These results align with our quantitative findings that users moved/copied more text, and added more words, article titles, and sections when using the OrgBox compared to the OrgDoc (Crescenzi et al., 2021). In our qualitative analysis, participants explained that the OrgBox was helpful because it prompted them to think about the information they collected. For example, creating a new box required them to generate a title and summarize the content. With multiple boxes side by side, participants could easily compare and identify gaps, which motivated them to read deeper and find additional information. These cognitive processes are crucial for learning and investigation during exploratory search (Marchionini, 2006; Wildemuth & Freund, 2012).

Our findings show that participants consciously integrated various activities, such as searching, reading, understanding, organizing, and saving during the exploratory search process. This aligns with previous research on the document triage process that identified intertwined patterns of selecting, reading, and organizing evaluated document relevance (Bae et al., 2006). Our results emphasize the importance of supporting searchers' cognitive behaviors, rather than just their search activities. The OrgBox tool facilitated mental activities such as creating and maintaining structures, analyzing and organizing information, and annotating intermediate perceptions and reflections. Previous work on document triage has considered the organizing of information at the document level (Badi et al., 2006; Hearst & Degler, 2013). Our work extends this to consider a tool that assists users to organize information at a lower level of granularity.

In our previous note-taking study, we observed that many participants took linear-structured notes during exploratory searches (Crescenzi et al., 2019). Our current findings suggest that with efficient tools (e.g. the OrgBox), users can generate complex structured notes even when their thoughts dynamically change. Some scaffolding tools in specific domains provide prompts or directions for users to pursue (e.g., (Stadtler & Bromme, 2008; Zhang & Quintana, 2012)). However, our findings demonstrate that even a tool that only provides structuring assistance (e.g., empty boxes) can benefit participants' cognition. These findings highlight the importance of supporting various cognitive activities throughout the *entire* exploratory search process, beyond just initiating exploration, making relevance judgement, and preserving search paths.

RQ2: Supporting metacognitive activities Our previous quantitative results (Crescenzi et al., 2021) demonstrated that participants rated the OrgBox significantly higher than the OrgDoc in terms of supporting cognitive and metacognitive activities including monitoring notes coverage, evaluating task progress, and understanding task topics (Crescenzi et al., 2021). The qualitative analysis presented in this paper complements the quantitative results in two ways. First, the qualitative findings provide details and explanations regarding *how* OrgBox features assisted users' cognitive and metacognitive behaviors. For example, the quantitative paper shows that the OrgBox helped participants in monitoring note coverage, which encompasses both breadth and depth. Our qualitative analysis revealed that the numbers of boxes

represented the number of aspects related to the task topic in the notes. Meanwhile, the dimensions (i.e., length) of each box indicate the volume of information gathered for a specific aspect.

Second, our qualitative findings illustrate that providing a visual overview of notes can benefit several types of metacognitive processes, including monitoring, planning, and evaluation. This expands on our previous quantitative results, where we did not evaluate the impact of visual assistance.

Our findings also confirm and extend previous work by elucidating why certain activities are crucial to support during exploratory search (e.g., saving search history, facilitating faceted exploration, saving metadata of retrieved information, organizing information, and taking notes (Morris, Ringel Morris, & Venolia, 2008; Donato, Bonchi, Chi, & Maarek, 2010; Guo et al., 2023; White & Roth, 2009)). In our work, we show how users' integrated their search behaviors and use of the interface features in support of their cognitive and metacognitive goals.

In the quantitative results, we did not find statistical significance of OrgBox for supporting planning. Our qualitative analysis suggests that this may due to 1) planning is generally hard even when people were provided with specific planning tools in a learning environment (Zhang & Quintana, 2012). It could be more challenging when they need to find information from an unfamiliar information topic area on the open web. 2) The influence of exploratory task design on users' search behavior, as established by previous research (Wildemuth & Freund, 2012). Given the study's purpose and experiment duration, participants were not asked to write a paper or produce any outcomes other than taking notes during the search process. This may have affected their planning concerning the extent of topic coverage and task completion.

RQ3: System usability evaluation RQ3 focuses on the usability issues of OrgBox compared to the OrgDoc baseline. Our previous quantitative analysis found that participants gave a higher *overall* evaluation rating to the OrgBox than to the OrgDoc (Crescenzi et al., 2021). However, our qualitative findings in this paper suggest that more participants preferred OrgDoc for its simplicity, familiarity, and flexibility of use (see Table 2).

While both OrgBox and OrgDoc exhibited common usability issues, a significant drawback of OrgBox was that participants felt OrgBox was more cumbersome to use due to its design features, which was not shown in the quantitative results. For example, participants reported that the automatic position adjustment of boxes (i.e., unexpected box movements) could disrupt their original organization and cause confusion. Additionally, redundant source tracking had a counterproductive effect by inducing more cognitive load for organizing information sources and the metacognitive load of tracking and monitoring. In spite of these disadvantages, the quantitative results of the significant differences between the interaction with OrgBox and OrgDoc (i.e., more text drags, added more words per drag, more article tiles, more sections, and more words in total) demonstrate participants made extensive use of OrgBox through the exploratory search process. Together, these findings suggest that the higher overall evaluation for the OrgBox indeed was more likely due to its unique features for support the exploratory search process, although its design needs to be improved further.

7 IMPLICATIONS

Our findings have implications for the design and evaluation of systems to support exploratory searches and other complex search processes.

Implications for Cognitive and Metacognitive Support in Search Context – Scaffolding has been widely used in the educational context to support students' cognition and metacognition. Our results indicate that providing a simple scaffolding tool (e.g. OrgBox) can help people engage more with the information they found on the web during an exploratory search. Tools for supporting create and maintain structures, dynamic organization, and annotation can

help users more engage in searching and thinking while reducing their mental workload for logistics (e.g., formatting). Allowing users to externalize their thoughts during search helps them keep track of their thinking process, monitor, and evaluate their approaches to the task. As a result, users' understanding of the task is no longer scattered but connected as a visible mental map guiding them to achieve their learning goals.

Implication for Complex Search Process Though this study mainly focuses on exploratory search, our findings suggest that tools like OrgBox can be applied to other complex search scenarios when users need to find information from different sources, collect different types of information, and conduct comparison, evaluation, and idea generation over time. For instance, Researchers found that complex tasks often require a great amount of cognitive and metacognitive activities as users need to find different types of information (e.g., factual, procedural information, opinions) (Choi, Ward, Li, Arguello, & Capra, 2019) and their solutions/paths to the outcomes are unclear due to open-ended task requirements like exploratory search (Byström & Järvelin, 1995; Campbell, 1988). The OrgBox tool can help distinguish different information types and topics by contrasting the information between boxes. Changing the arrangement of boxes help people compare potential paths (e.g., pros vs. cons) to decide their search strategies. The special feature of boxes can provide cues to help users to continue interrupted tasks.

Implication for Search Assistant Tools' Evaluation Finally, our study suggests that the evaluation of tools for supporting exploratory search should take supporting users' cognitive and metacognitive activities into account. Unlike fact-finding tasks, exploratory search focus on both the exploring process and the learning outcome. Previous evaluations of search assistant tools and other learning tools mainly focused on comparing users' search activities and their task outcomes (e.g., knowledge tests) under different conditions. Little attention has been paid to the support for the mental process. Our results indicate that the tools can be evaluated based on their efficiency and effectiveness in supporting cognitive and metacognitive activities during the search process.

8 CONCLUSION

In this study, we present a design of a novel search assistance tool called *OrgBox* for helping searchers save, annotate, organize, and synthesize the information found through the online search process. To evaluate how the *OrgBox* supports the information search process and understand users' perceptions about the benefits and challenges of using *OrgBox*, we conducted a controlled, within-subjects user study (via Zoom) with 24 participants. Through the study, we compared the *OrgBox* tool to a baseline condition called the *OrgDoc* that mimicked the features found in rich-text editors. Participants used each tool to complete a different exploratory search task. They completed questionnaires after each task with open-ended questions about their perceptions of the search and the *OrgTool* systems.

Overall, participants reported that the *OrgBox* provided significant support for different cognitive and metacognitive activities through their search processes. The *OrgBox* helped them plan and initiate the search by focusing on various aspects related to the task. The box scaffolding helped them organize and synthesize the collected information to prepare for the final task outcomes within the search context. The visual appearance of boxes is beneficial for evaluating and monitoring the search process and task progress. The features of simplicity, familiarity, and flexibility make the *OrgTool* system easy to use, whereas there are still some restrictions of the *OrgBox* that could be improved in the future. Our findings have implications for designing search systems and tools to support cognitive and metacognitive activities during complex search tasks.

NOTES

¹More details about the design of OrgBox can be found in Ward and Capra (2021).

²These results were presented in Crescenzi et al. (2021), and are summarized here to provide important context for understanding and interpreting our qualitative results.

³The exploratory factor analysis results are available at: <https://doi.org/10.17615/c7b0-2t77>.

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